Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 1 of 7 · Question 2 part b · 6 marks

2024 JUN · Paper 01 · find part b in this mark scheme extract

2 (a)	(i)	• Y axis label and units; • X axis bars identified; • scale; • plots;	line graph score mpt 1-3	(4)
	(ii)	1;		(1)
(b)		In the following order: • lag; • exponential; • equal to; • death; • less than; • dying;		(6)

Total for Question 2 = 11 Marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 2 of 7 · Question 4 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

4 (a) (i)		A (psychrophiles); B is incorrect as the temperature of a fridge is not between 12 and 45 degrees C is incorrect as the temperature range is too high for a fridge D is incorrect as the temperature range is too high for a fridge		1
	(ii)	increase in growth/maximum growth; (near) optimum/best temperature;	Allow optimum growth.	1 1
(b)	(i)	measure diameter/area of inhibition/of the clear zone/area of no bacterial growth; greater the diameter, the more effective the antibiotic;	Allow observe size of the clear zone.	2
	(ii)	wash hands/work surface after carrying out procedure; reduces transfer of bacteria; Heat inoculating loop/spreader; To kill bacteria/to prevent contamination; do not open dish following incubation/seal the Petri dish (prior to incubation); reduces risk of bacteria being transferred to student/infection; do not incubate over 25°C; prevents growth of pathogenic bacteria; Sterilise/clean all equipment before/after use; Reduces cross contamination; Seal petri dish/do not open lid of petri dish too far/reduce exposure of contents of dish to air; Reduces contamination by airborne bacteria/to prevent bacteria escaping;	Max 4 if no explanation given.	6 max
			Total question 4 = 11 marks	

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 2 of 7 · Question 4 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 3 of 7 · Question 5 part b.ii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
5	(a)	(i)	• use people from a range of ages; • hearing test using high frequency sounds; • control other variables e.g. sex, other health issues/same distance from sound source;		3
		(ii)	exposure to loud noises/genetic cause/infection		1
	(b)	(i)	• vibrations of the eardrum not transferred to the ossicles; • vibrations are not transferred to the cochlea; • hair cells are not stimulated/impulses not generated/not transferred to the auditory nerve/brain;		3
		(ii)	Two from: • conditions are moist/warm; • ideal conditions for bacterial growth/to multiply; • pathogens/bacteria more likely to get trapped;		2
		(iii)	Three from: • semi-circular canals; • contain fluid (in three tubes); • as the head moves so does the fluid; • transmits the information to the brain/cerebellum; • (cerebellum)co-ordinates a response to maintain balance/control balance;		3

Total 12 marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 3 of 7 · Question 5 part b.ii · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 4 of 7 · Question 4 part a.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

4	(a)	(i)	A (psychrophiles); B is incorrect as the temperature of a fridge is not between 12 and 45 degrees C is incorrect as the temperature range is too high for a fridge D is incorrect as the temperature range is too high for a fridge		1
		(ii)	increase in growth/maximum growth; (near) optimum/best temperature;	Allow optimum growth.	1 1
	(b)	(i)	measure diameter/area of inhibition/of the clear zone/area of no bacterial growth; greater the diameter, the more effective the antibiotic;	Allow observe size of the clear zone.	2
		(ii)	wash hands/work surface after carrying out procedure; reduces transfer of bacteria; Heat inoculating loop/spreader; To kill bacteria/to prevent contamination; do not open dish following incubation/seal the Petri dish (prior to incubation); reduces risk of bacteria being transferred to student/infection; do not incubate over 25°C; prevents growth of pathogenic bacteria; Sterilise/clean all equipment before/after use; Reduces cross contamination; Seal petri dish/do not open lid of petri dish too far/reduce exposure of contents of dish to air; Reduces contamination by airborne bacteria/to prevent bacteria escaping;	Max 4 if no explanation given.	6 max

Total question 4 = 11 marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 4 of 7 · Question 4 part a.i · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 5 of 7 · Question 1 part a · 1 mark

2020 JAN · Paper 01 · find part a in this mark scheme extract

Question number				
1	(a)	**The only correct answer is C** to slow down the growth of bacteria *A is incorrect because taste is personal preference* *B is incorrect because it does not kill the bacteria* *D is incorrect because the appearance is not affected*		1
	(b) (i)	protein		1
	(ii)	**The only correct answer is D** it does not contain energy from fat *A is incorrect because vitamins have no effect on body mass* *B is incorrect because cholesterol has no effect on body mass* *C is incorrect because it does contain energy from carbohydrates*		1
	(c)	• add Benedict's solution (to a sample of yogurt) (1) • heat gently (1) • brick red colour indicates the presence of sugar (1)	R if adding to glucose/reducing sugar Allow yellow, green, orange	3

Total for Question 1 = 6 marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 6 of 7 · Question 6 part a.iv · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

Number					
6	(a)	(i)	18-12; 6;	accept 17.8/17.9	2
		(ii)	Two from: • volume of milk; • volume of the starter culture/live yoghurt; • type of starter culture • type of milk; • type of live yoghurt; • mixing;	reject 'amount'	2
		(iii)	• repeat the investigation at more temperatures; • between 20°C and 60°C;		2
		(iv)	• heated to kill (unwanted) bacteria in the milk; • cooled so that the bacteria in the starter culture are not killed;		2
	(b)		Any two from: • pH 4.4 inhibits the bacteria/bacterial growth; • denatures enzymes; • changes the shape of the active sites • lactose cannot bind to the enzyme/no enzyme-substrate complexes formed;		2
	(c)	(i)	Any two from: • the beads can be re-used; • product doesn't need to be treated to remove the enzyme; • large surface area to increase the rate of reaction; • enzyme more temperature stable;		2
		(ii)	• long/villi/microvilli to increase surface area; • concentration gradient maintained with blood flow; • short diffusion distance;		3

Total 15 marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 6 of 7 · Question 6 part a.iv · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain bacterial growth conditions and how food preservation slows it

Answer 7 of 7 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

2 (a)	(i)	• Y axis label and units; • X axis bars identified; • scale; • plots;	line graph score mpt 1-3	(4)
	(ii)	1;		(1)
(b)		In the following order: • lag; • exponential; • equal to; • death; • less than; • dying;		(6)

Total for Question 2 = 11 Marks